

SUMMARY

Ph.D. candidate developing turbulent flow control methods using CFD, numerical modeling, and high-performance computing to optimize advanced aircraft surface performance. Seeking CFD focused internship opportunities.

EDUCATION

Arizona State University, Tempe, AZ, USA 01/2023 – 05/2027 (Anticipated)

Ph.D. Mechanical Engineering, GPA: 4.0/4.0

Research Focus: CFD, Turbulent Flow Control, Spectral Methods, High Performance Computing.

Arizona State University, Tempe, AZ, USA 01/2023 – 12/2025

Master of Science Mechanical Engineering, GPA: 4.0/4.0

RELEVANT EXPERIENCE

Arizona State University 01/2023 – Present

Ph.D. Researcher developing computational methods for turbulent flow optimization with applications in aerospace and thermal-fluid systems:

- Developed frequency-tuned flow control approaches to wall-transpiration control methods using Fourier transforms and averaging in spectral code Nek5000 and obtained upto 9% drag reduction [2].
- Developed novel flow control methods based on wall information in turbulent channel flow and looked at drag reduction problems, achieving upto 12% drag reductions; DNS in open-source CFD code, Nek5000 [3].
- Studied separated flow physics on a backward-facing step using frequency analysis, and modal decomposition to get the frequencies for flow control using meta-materials; validated with experimental collaboration [1].
- Exploring wall-information method for flow separation control on a BFS using high performance computing.
- Exploring ML for wall pressure prediction for flow control with CNNs in the advanced spectral code, NekRS.

IOE, Tribhuvan University 11/2021 – 10/2022

Teaching Assistant

- Assisted ME 431: Engineering Drawings I; taught classes and labs on engineering drawing and CAD modeling.
- Assisted SH 601: Numerical Methods; led labs on numerical methods using C and Python.

Nepal Electricity Authority Engineering Company 05/2021 – 08/2021

Mechanical Engineering Intern

- Designed hydro-mechanical components for 480 MW Phukot Hydropower Plant, contributing to feasibility study for large-scale power generation in remote mountainous region of Karnali River.
- Performed power plant layout design integrating hydro-mechanical components, turbines, generators, and cooling systems, ensuring optimal thermal management and spatial arrangement within facility constraints.
- Created technical drawings and specifications for hydro-mechanical systems using AutoCAD and SolidWorks, coordinating with civil and electrical engineering teams for integrated powerplant design.

SKILLS

Softwares: Nek5000/RS, OpenFOAM, ANSYS Fluent, Paraview, Solidworks, AutoCAD.

Technical: Finite Difference/Volume/Element, and Spectral methods, HPC, Machine learning.

Programming Languages: Python, MATLAB, C, C++, FORTRAN. **OS:** Linux/Unix systems.

SELECTED PUBLICATIONS

- [1] **A. Subedi**, and Y. Peet. *Separated Flow Physics Using Frequency Analysis and Modal Decomposition*. TSFP, 2026.
- [2] **A. Subedi**, and Y. Peet. *Frequency-Tuned Approaches to Wall Transpiration Control*. AIAA SciTech Forum, 2025.
- [3] **A. Subedi**, and Y. Peet. *Turbulent Flow Control using Wall Sensing*. AIAA Aviation Forum and Ascend, 2024.
- [4] A. Pandey, M. Timsina, **A. Subedi**, S. Gautam, K. Darlami, & S. Bhattarai. *Modal Decomposition of Flow During Dynamic Stall in NACA 4412 Airfoil Using Proper Orthogonal Decomposition*. AIAA SciTech Forum, 2025.

Full list on Google Scholar.

COURSEWORKS & ACTIVITIES

Courseworks: Fluid Mechanics, Thermodynamics, Heat Transfer, Turbulence, Aircraft Propulsion, Computational Fluid Dynamics, HPC, PDEs, Spectral Element Methods, Linear Algebra, Machine Learning.

Awards: Experiential Learning Grant, GPSA Travel Grant, Graduate College Travel Grant. 2024 and 2026

Affiliations: Member of AIAA and American Physical Society DFD. 2024 - Present